

LingoLearn: Live the language through Immersive Augmented Reality Cultural Scenarios and Practice

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Contents

1	Introduction	1
1.1	Problem Statement	2
1.2	Scope	2
1.3	Modules	3
1.3.1	User Interface (UI) Module	3
1.3.2	Progress Tracking and Performance Analytics Module	3
1.3.3	Scenario Based Assessment Module.	4
1.3.4	Module of an Augmented Reality AR Environment	4
1.3.5	AI Language Practitioner Module	4
1.3.6	Pronunciation Feedback Module	5
1.3.7	Content Management Module	5
1.3.8	Backend and Database Management Module	5
1.4	User Classes and Characteristics	6
2	Project Requirements	7
2.1	Use-case	7
2.1.1	Extended Use Cases	12
2.2	Functional Requirements	19
2.3	Non-Functional Requirements	22
2.3.1	Reliability	22
2.3.2	Usability	22
2.3.3	Performance	22
2.3.4	Maintainability	22
2.3.5	Scalability	22
2.3.6	Compatibility	23
2.3.7	User Feedback	23
2.3.8	Environmental Impact	23
2.4	Domain Model	24
3	System Overview	25
3.1	Architectural Design	25
3.2	Design Models	25
3.2.1	Data Flow Diagrams	25

CONTENTS

3.2.2	System Level Sequence Diagrams:	39
3.2.3	State Transition Diagram	49
3.3	Data Design	50
3.3.1	Entity Relationship Diagram:	50
References		51

List of Figures

2.1	UseCase Diagram	8
2.2	Domain Model	24
3.1	Architectural Design Diagram	25
3.2	Level 0: Context Diagram	26
3.3	Level 1 Diagram	27
3.4	Level 2 Diagram	28
3.5	Activity Diagram 1:Sign Up	29
3.6	Activity Diagram 2:View Results	30
3.7	Activity Diagram 3:View Streaks	31
3.8	Activity Diagram 4:Take test	32
3.9	Activity Diagram 5:Interact with Learning Assistant	33
3.10	Activity Diagram 6:Take AR Scenario-Based Assessment	34
3.11	Activity Diagram 7:Generate Certificate	35
3.12	Activity Diagram 8:take lecture	36
3.13	Activity Diagram 9:Manage Profile	37
3.14	Activity Diagram 10:Register Course	38
3.15	SSD 1:Sign Up	39
3.16	SSD 2:Manage Profile	40
3.17	SSD 3:Register Course	41
3.18	SSD 4:Take Lecture	42
3.19	SSD 5:Take Test	43
3.20	SSD 6:Take Assessment	44
3.21	SSD 7:View results	45
3.22	SSD 8:View Streaks	46
3.23	SSD 9:Interact with AI Assistant	47
3.24	SSD 10:Generate Certificate	48
3.25	State Transition Diagram	49
3.26	Entity Relationship Diagram	50

List of Tables

2.1	Use Case: 1	9
2.2	Use Case: 2	9
2.3	Use Case: 3	9
2.4	Use Case: 4	10
2.5	Use Case: 5	10
2.6	Use Case: 6	10
2.7	Use Case: 7	10
2.8	Use Case: 8	10
2.9	Use Case: 9	10
2.10	Use Case: 10	11
2.11	Use Case: 11	11
2.12	Use Case: 12	11
2.13	Use Case: 01	12
2.14	Use Case: 02	13
2.15	Use Case:03	13
2.16	Use Case: 04	14
2.17	Use Case: 05	14
2.18	Use Case: 06	15
2.19	Use Case: 07	15
2.20	Use Case: 08	16
2.21	Use Case: 09	16
2.22	Use Case: 10	17
2.23	Use Case: 11	17
2.24	Use Case: 12	18
2.25	Functional Requirements	21

Chapter 1

Introduction

The aim of this project proposal document is to describe in detail the conception and development of an immersive language learning platform utilizing Augmented Reality (AR) and Artificial Intelligence (AI). The document contains the objectives, goals, tools, and technologies of the project as well as the problem statement which defines the issues this platform seeks to solve. This proposal is useful to the stakeholders since it describes in detail what the platform will do and the processes that are going to be used to develop the platform.

Language is still one of the last standing barriers when it comes to global communication, especially in education as well as professional institutions. It's often boring to practice many traditional methods of learning languages, and conversational tools are limited. Plus, live sessions with native speakers are pricey and difficult to manage; there is little flexibility when it comes to this option. Increasingly, AR and AI technologies present new opportunities to make language learning more interactive, immersive, and available than ever before.

This project suggests building a mobile-based AR app that offers an immersive, scenario-driven learning setup supported by AI language companions for conversational practice on the fly. With its characteristics, the platform scope searches for current gaps in existing language learning methods such as trying to create a real environment in AR and using AI for effective phrase pronunciation, vocabulary, and fluency. This application focuses on learners, corporate people, the workforce, and travelers and they will all have an interesting and effective way of learning a language with the available languages including Urdu, English, and German.

The project seeks to transform language acquisition processes using Augmented Reality technology to create an environment where learners can engage in practical activities and work with Artificial Intelligence to improve feedback, interaction, and tracking of progress learning through language. The scope of this document is to describe the sys-

tem's boundaries, goals, objectives, and context while ensuring that the development is user-centered and diverse.

1.1 Problem Statement

Language is still a considerable barrier that hinders communication in education, business and daily life[1], especially for students who apply for scholarships in foreign countries where language proficiency is a requirement. Even with an abundance of language learning apps out there, many are boring and unengaging, making it hard for people to stay motivated when they're trying to learn a new language. These apps also lack real-time scenario practice or on-the-fly pronunciation feedback, which is so essential for learning a language. Live sessions with native speakers can be expensive or difficult to schedule due to timezone issues, reducing one of the greatest advantages — flexibility. Our system helps overcome these shortcomings by creating a sensory and cognitively immersive language learning experience with AR, on top of AI. This provides practice for real-life situations, instant pronunciation feedback, and communication with a virtual language partner in the learner's own time. We want to evolve from the shortcomings of traditional apps and provide an engaging, affordable, practical alternative that leads users back on their journey toward achieving language fluency without class or financial time constraints. [2]

1.2 Scope

This project involves the creation of a mobile language learning app that will leverage Augmented Reality (AR) and Artificial Intelligence (AI). Key features of the app include a Mobile Augmented Reality (MAR), which takes users into 3D real-world situations and immerses them in cultural language contexts for learning. A user can use his phone and explore an AR environment which improves learning vastly, making the experience more intuitive and unforgettable. An AI-powered language accompaniment is a personalized tutor offering real-time guidance and natural practice conversations. The app comes with enhanced speech recognition that gives feedback on your pronunciation, so you can work on speaking practice in the best possible way. It also includes real-world scenario-based final assessments to evaluate whether the learner can put their language skills to use like an example of ordering food or traveling through a foreign market. Track your progress and performance with analytics so that you stay driven throughout the learning process, instilling a sense of streaks along with personalized feedback. Students business professionals, working employees, and even tourists- all can achieve their requirements with this platform. It starts with users in Pakistan, the UK, and Germany — but there are plans

to roll it out more broadly. The available languages are English, German, and Urdu more to come later. Our application is built on Augmented Reality for interactive learning and Artificial Intelligence is used for personalization — via mobile app.

1.3 Modules

Following are the modules that are included in our project:

1.3.1 User Interface (UI) Module

Develop the quickly comprehensible and manipulable online front-end and applications with the minimum accessibility barriers for browsing and exploiting your attributes.

Features:

1. Home screen with links for navigation engaging lessons, tests, and progress.
2. Lesson screen with the AR scenes and interaction with AI bots.
3. Screens for conducting the pronunciation test as well as scenario type assessment screens.
4. Screens that show feedback and analytics concerning the user's performance.

1.3.2 Progress Tracking and Performance Analytics Module

Monitor user's language progress and offer language learning analytics from data collected.

Features:

1. Aggregate information on user engagement, test results, as well as learning achievements.
2. Performance analytics including metrics and insights based on the user.
3. System Dashboard to demonstrate user completion and progress as well as possible deficits.

1.3.3 Scenario Based Assessment Module.

Define language problems with the learner, and use an approach where the user learns by doing and acts out real-life situations by speaking the language in the assessment.

Features:

1. Create scenarios that replicate and expose real-world interaction.
2. Create and implement a logic that will assess the user's answers and give that answer a mark.
3. Vary the complexity of the scenarios according to the ability of the user.

1.3.4 Module of an Augmented Reality AR Environment

Design extraordinary AR environments that incorporate the cultural and contextual factors related to the target language being learnt.

Features:

1. Provide a virtual world with the alternating cultures for a proper language acquisition.
2. Quickly change the environment according to the situation.

1.3.5 AI Language Practitioner Module

Assist learners through an AI bot who provides instant support on the learned language and feedback including comments on the accuracy of words or sentences used in conversations.

Features:

1. Interact by speaking which will involve the use of speech recognition.
2. Responds using AI well heavens accuracy and relevance of speed and comprehension.
3. Helps one speak the language by simulating the necessary accent.

1.3.6 Pronunciation Feedback Module

Assist users improving themselves in pronouncing certain notions thanks to a detailed response and feedback with intervals.

Features:

1. Conduct a speech analysis to determine the level of the pronunciation efforts.
2. Render a feedback immediately with ways to get better.

Query Resolution and Support Module

Assist users in real-time while they learn by using the Android-assisted Support system.

Features:

1. Resolve user queries related to their learning experience on the spot.
2. Respond to user questions by generating automatic predetermined answers.

1.3.7 Content Management Module

Supervise lesson and test development, as well as editing.

Features:

1. Design and change lesson and evaluation types.
2. Compile lesson plans according to the skills and knowledge levels of pupils.
3. Make logs of content additions and changes for system maintenance.

1.3.8 Backend and Database Management Module

Conduct strict and secure data handling and offer management of backend processes of the platform.

Features:

1. Administer user accounts, progress data and individual preferences and settings.
2. Administer lessons and assessments concepts and AR scene data.
3. Provision of data security facilities.

1.4 User Classes and Characteristics

User class	Description
Students	Students are individuals who are registered within some organization, particularly institutions of learning, and are interested in language acquisition. The application might be used in enhancing one's studies or even learning personally. It is anticipated that students will always turn to the application on a regular basis for lessons, pronunciation practice, and AR-based scenario exercises. Numerous learners are likely to utilize the application for examinations or for reasons such as scholarship requirements, and they expect clear progress tracking. They would use AR rich modules and AI teaching avatar to enhance their experience.
Tourists	Tourists are occasional learners who want to grasp basic conversational skills in a foreign language for travel purposes. They will typically use this mobile application to learn phrases from the AI tutor and will be more inclined to use the app to prepare for specific situations like ordering food, asking for directions, or navigating foreign markets. For tourists, the focus of interactions through the app will be the AR environments rather than learning languages in other cultures.
Staff	Staff members are people who will seek to embark on language acquisition as one way of improving for better job performance, promotions or enhancement of workplace integration. These users may be using the app in the office where they try to incorporate learning while going about their day to day tasks.
Language Enthusiasts	These users are highly motivated who intend to acquire proficiency in various languages and also further their linguistic skills in one language. Such users will make use of all the functionalities which are given to them including the advanced AR module and in depth lessons. Enthusiasts can be expected to change languages quite often, and due to that, they will want to be able to switch languages seamlessly. This class of users will also use AI companion for personalized learning experiences and they expect the system to be reliable and offer challenging tasks and tests, as they are likely to be committed persistent users.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case

Use-case Diagram

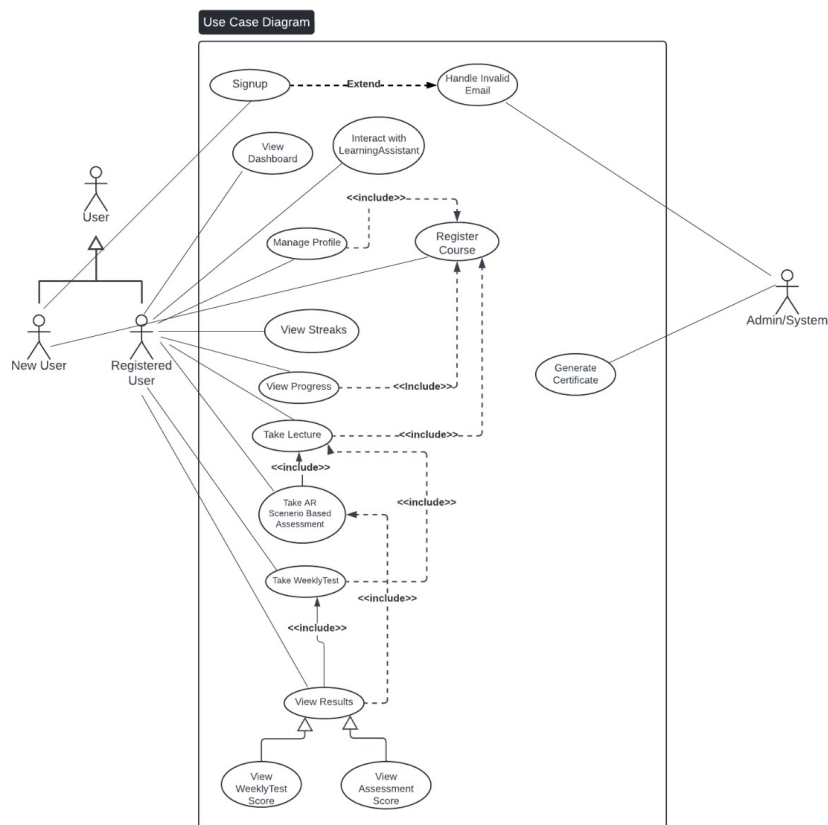


Figure 2.1: UseCase Diagram

In our case of this mobile language learning app, we have decided to go for the Use Case. This technique works perfectly when detailing an interactive, user's utility-driven app like ours, where the clients deal with AR environments, and AI language supporters, and engage in lectures.

Use Case Name	User Signup
Actors	User
Type	Primary
Description	The user registers for the application by providing the required details

Table 2.1: Use Case: 1

Use Case Name	View Dashboard
Actors	User
Type	Primary
Description	Displays registered courses, progress, streaks, and navigation options.

Table 2.2: Use Case: 2

Use Case Name	Manage Profile
Actors	User
Type	Primary
Description	The user can update their personal details and preferences.

Table 2.3: Use Case: 3

Use Case Name	Register Course
Actors	User
Type	Primary
Description	The user selects a language course and their proficiency level to enroll.

Table 2.4: Use Case: 4

Use Case Name	Take a Lecture
Actors	User
Type	Primary
Description	The user reviews previously completed lectures and takes the next available one.

Table 2.5: Use Case: 5

Use Case Name	Take a Test
Actors	User
Type	Primary
Description	The user completes a test after finishing a week of lectures.

Table 2.6: Use Case: 6

Use Case Name	Conduct AR Scenario-Based Assessment
Actors	User
Type	Primary
Description	The user completes AR assessments halfway through and at the end of the course.

Table 2.7: Use Case: 7

Use Case Name	Interact with Learning Assistant
Actors	User
Type	Primary
Description	The user interacts with the AI avatar to ask language-related questions.

Table 2.8: Use Case: 8

Use Case Name	View Progress
Actors	User
Type	Primary
Description	The user can see their overall progress in the course, including how many lectures are left and the week they are on.

Table 2.9: Use Case: 9

Use Case Name	View Results
Actors	User
Type	Primary
Description	The user views results from all the tests and AR assessments.

Table 2.10: Use Case: 10

Use Case Name	View Streaks
Actors	User
Type	Primary
Description	The user views the number of consecutive days they have used the app.

Table 2.11: Use Case: 11

Use Case Name	Issue certificate
Actors	System
Type	Primary
Description	The system generates a digital certificate for the user when they complete the whole course on a specific criteria

Table 2.12: Use Case: 12

2.1.1 Extended Use Cases

Use Case ID	Uc-01	
Use Case Name	User Signup	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The app is installed	
Success Guarantee (Post Condition)	The User is successfully registered	
Main Success Scenario	Actor Action	System Response
	User opens the app and selects "Sign Up".	
		The system asks the user to enter required details.
	User enters the required details.	
		The system verifies the inputted data.
		The system creates a new user and sends a confirmation mail.
	User clicks the verification link in the email.	
		The system confirms the account is verified and completes registration.
Special Requirements	The user must not be already registered; otherwise, they will see an "already registered" message.	
Technology and Data Variations List	Internet connection	
Frequency of Occurrence	Every time a new user wants to use the app	

Table 2.13: Use Case: 01

Use Case ID	Uc-02	
Use Case Name	View Dashboard	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user is logged in and has been registered to atleast 1 course	
Success Guarantee (Post Condition)	The Dashboard successfully displays the User's registered courses, progress and streaks	
Main Success Scenario	Actor Action	System Response
	User Logs in the app	
		The system fetches the user data and the courses they are registered in.
		The dashboard is displayed, showing information such as progress, courses registered etc
	User can go to the different option on the dashboard	
Special Requirements	The dashboard should update dynamically	
Technology and Data Variations List	Internet connection	
Frequency of Occurrence	Everytime the user logs in or navigates to dashboard.	

Table 2.14: Use Case: 02

Use Case ID	Uc-03	
Use Case Name	Manage Profile	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user is logged in.	
Success Guarantee (Post Condition)	The User profile is updated dynamically and the changes are visible after saving.	
Main Success Scenario	Actor Action	System Response
	User selects "Manage Profile" from the navigation bar.	
		The system displays the current information of the user.
	The User updates the profile information.	
		The system validates the information.
		System notifies the user that the changes have been saved.
Special Requirements	The system should handle the information securely.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Everytime the user wants to update their information.	

Table 2.15: Use Case:03

2. Project Requirements

Use Case ID	Uc-04	
Use Case Name	Register Course	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user is logged in.	
Success Guarantee (Post Condition)	The user is successfully registered in a course and the dashboard is updated accordingly.	
Main Success Scenario	Actor Action	System Response
	User selects "Register course" from the navigation bar.	
		The system displays the list of all the available courses that can be registered.
	User selects the language and their proficiency level.	
		The system confirms course registration and updates the user's dashboard
		System notifies the user that the course has been registered.
Special Requirements	The system should offer the course according to the proficiency of the user.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Every Time the user wants to register in to a new course.	

Table 2.16: Use Case: 04

Use Case ID	Uc-05	
Use Case Name	Take a Lecture	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user is logged in and is also enrolled in a course	
Success Guarantee (Post Condition)	The user completes the lecture and the dashboard is updated automatically.	
Main Success Scenario	Actor Action	System Response
	User selects "Take Lecture" from the navigation bar.	
		The system displays the list of all completed and pending lectures.
	User selects the next lecture	
		The system displays the content of the lecture.
		The system marks the lecture as completed and updates the dashboard.
Special Requirements	The lecture must be accessible and should follow a proper outline.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Every Time the user wants to take a lecture.	

Table 2.17: Use Case: 05

Use Case ID	Uc-06	
Use Case Name	Take a Test	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user has complete the number of lectures required to take a test.	
Success Guarantee (Post Condition)	The user completes the test and gets feedback as well as score are displayed on the dashboard	
Main Success Scenario	Actor Action	System Response
	User opens the lesson from navigation bar	
		System displays all the lectures of the upcoming week along with the tests available
	User selects the upcoming test	
		System displays the test
	User does the required tasks	
		System updates the results and progress and dashboard is updated.
Special Requirements	Tests are to be taken after a specific period of time such as after a week or so.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	At the end of each week or specific amount of time	

Table 2.18: Use Case: 06

Use Case ID	Uc-07	
Use Case Name	Conduct AR Scenario-Based Assessment	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user has completed the amount lectures and test specified by the admin.	
Success Guarantee (Post Condition)	The user has completed the AR Assessment and has been evaluated.	
Main Success Scenario	Actor Action	System Response
	User opens the lesson from navigation bar	
		System displays all the lectures of the upcoming week along with the tests available
	User selects the upcoming Assessment	
		The system then instantiate the AR environment
	User interacts with the environment to complete the assessment	
		System evaluates the user's performance in the Scenario and gives feedback on the performance and updates the user dashboard
Special Requirements	The app should provide realistic AR scenario for the user	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	At a specific milestone such as mid-term, final assessment.	

Table 2.19: Use Case: 07

2. Project Requirements

Use Case ID	Uc-08	
Use Case Name	Interact with Learning Assistant	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user is logged in and is actively registered in a course	
Success Guarantee (Post Condition)	The user is able to interact with the AI bot in AR environment and gets accurate response from the bot.	
Main Success Scenario	Actor Action	System Response
	User selects "Learning Assistant" from dashboard	
		The system displays an AI bot in AR environment which initiates the conversation.
	The user then interacts with the AI bot to practice.	
		The system then responds accordingly and help the user improve grammar/ pronunciation.
		System remains in conversation with counter questions.
Special Requirements	The app should provide realistic AR scenario for the user	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Anytime the user wants additional help without taking lessons or wants to practice in freestyle with restriction.	

Table 2.20: Use Case: 08

Use Case ID	Uc-09	
Use Case Name	View Results	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The User has complete atleast one test or assessment.	
Success Guarantee (Post Condition)	The User can see their progress and scores of all test/assessment they have completed	
Main Success Scenario	Actor Action	System Response
	User selects "View Results" from the navigation bar.	
		The system retrieves all the test/ assessment results of the user.
	The user sees detailed results with score and feedback given on the test.	
	Users can review their progress and compare results over time.	
Special Requirements	The Results should be of the user that requested and should be displayed in a user-friendly manner.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Anytime user completes a test or assessment.	

Table 2.21: Use Case: 09

Use Case ID	Uc-10	
Use Case Name	View Streaks	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user has logged consecutively on the app for multiple days.	
Success Guarantee (Post Condition)	The users streaks are displayed.	
Main Success Scenario	Actor Action	System Response
	User logs in the app	
		The system checks the user activity and calculates the streaks.
	User views streaks history	The system updates the user streaks everyday they login.
Special Requirements	User is given notification to maintain streaks.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Every day the user logs in the app	

Table 2.22: Use Case: 10

Use Case ID	Uc-11	
Use Case Name	View Progress	
Scope	Lingo Learn	
Level	User-Level	
Primary Actor(s)	User	
Stakeholders and interests	User, App Admin	
Pre-Conditions	The user is logged in and is registered in atleast one course.	
Success Guarantee (Post Condition)	The user is able to see their progress such as lessons taken, test scores	
Main Success Scenario	Actor Action	System Response
	User selects "View Progress" from the dashboard.	
		The system displays the current status of the user.
	User views progress of each lesson/ tests/Assessments in detail	
	The user exist the progress view	
Special Requirements	The system updates the progress dynamically.	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Every time the user wants to check their progress.	

Table 2.23: Use Case: 11

2. Project Requirements

Use Case ID	Uc-12	
Use Case Name	Issue certificate	
Scope	Lingo Learn	
Level	System-Level	
Primary Actor(s)	System	
Stakeholders and interests	User, System	
Pre-Conditions	The user has completed the whole course	
Success Guarantee (Post Condition)	The user is able to see and download certificate.	
Main Success Scenario	Actor Action	System Response
	After the final Assessment user selects generate Certificate	
		The system then verifies the criteria.
	The user downloads and exits.	when Verified the certificate is displayed
Special Requirements	The certificate should have the User's name	
Technology and Data Variations List	Internet connection.	
Frequency of Occurrence	Everytime a course has been completed by the user	

Table 2.24: Use Case: 12

2.2 Functional Requirements

Use Case	ID	Functional Requirement
User Signup	FR-001	The system shall have an option for users to create an account by filling in the details such as email address and password.
	FR-002	The system shall notify the user with an error if the email address is already registered in the system during signup.
	FR-003	The system shall create a one-time password during the signing process when the user registers, and it shall be sent to the email supplied during signup.
	FR-004	The system shall confirm the user's OTP input against the one sent to their email and then confirm the user's email upon successful verification.
User Login	FR-005	The system shall enable the users to log in through their noted email addresses and their created passwords.
	FR-006	The system shall show an error when the user enters incorrect login details while logging into the system.
Dashboard	FR-007	The system shall display a dashboard with all the user's registered courses and the course progression once the user has successfully logged in.
	FR-008	The system shall calculate and present the total completed courses and the current week and day of the active courses on the dashboard.
	FR-009	The system shall display numbers of streaks on the dashboard and the numbers will be incremented for every day that a user accesses the app.
Take a Lecture	FR-010	The system shall present an interface that shall contain a button on the dashboard which shall redirect the determined user to the available lectures.
	FR-011	The system should provide the users with a list of lectures comprehensively along with the lectures' titles and the completion date.
	FR-012	The system shall provide an option for the users to speak to the AI tutor during lectures to support real-time communication and assistance.
	FR-013	The system shall offer an AI avatar tutor to users in an Augmented Reality (AR) setting when taking a lecture, enhancing the experience by making it more interactive and participatory.

View Streaks	FR-014	The system shall provide the user with the information regarding the streak count on the dashboard.
	FR-015	The system should give users the option of clicking on the streak section to view the streak history.
View Progress	FR-016	The system shall display the overall learning progress of a user on the dashboard.
	FR-017	The system should allow the users to see their progress through the use of diagrams or charts.
Manage Profile	FR-018	The system shall allow the users to navigate to the section, “Manage Profile” through the navigation bar.
	FR-019	The system shall enable the users to change their personal information on the system under the section, Manage Profile.
Register Course	FR-020	The system shall offer users the choice to enroll in a certain category through language and then a proficiency level option for a new course.
	FR-021	The system should display a confirmation message upon successful enrollment of the course.
Interact with the Learning Assistant	FR-022	The system should have a conversational agent who works as a virtual teacher and answers users’ questions regarding the process of learning a language.
	FR-023	The system should allow the users to give input to the AI learning assistant avatar by speaking in a human voice and processing this input for appropriate responses.
View Results	FR-024	The system shall allow the users to navigate to the Results section through the navigation bar to see all the test and AR scenario-based assessment results.
	FR-025	The results screen of this system shall also display the user’s pronunciation accuracy over a certain period with the other test results.
Change Settings	FR-026	The system shall provide users the option to change the app settings through the navigation bar.
	FR-027	The system should store the user preferences and user settings on submission for later use.

Take tests	FR-028	The system must allow the users to perform scenario-based learning activities in the AR spaces where they will simulate communication and tasks one would do in real life.
	FR-029	The system shall provide clear instructions and target objectives of every scenario-based assessment in detail before the user commences the assessment.
	FR-030	The system shall also accept answers in the form of spoken words/sentences during weekly tests enabling the system to evaluate the user's ability to pronounce the right words in addition to how correct the answers are.
	FR-031	The system shall provide immediate feedback on user performance after completing the assessment, highlighting strengths and areas for improvement.
	FR-032	The system shall provide feedback on user performance in tests promptly highlighting pronunciation accuracy.
Issue Certificate	FR-033	The system should provide users who have completed the requisite number of lectures and assessments with the completion certificate.

Table 2.25: Functional Requirements

2.3 Non-Functional Requirements

Following are the non functional requirements of our Project "Lingo Learn: Live the language through Immersive Augmented Reality Cultural Scenarios and Practice" :

2.3.1 Reliability

- The system must be available for users 99.9 percent of the time.
- There should be ways to recover data if there is any failure of device.

2.3.2 Usability

- The interface should be user friendly so that user of any age can navigate through the app easily.
- There should be a manual/ tutorial to help users using the app for the first time to understand the app working.

2.3.3 Performance

- The landing page should not take more the 3 seconds to load.
- The AR experience should be smooth in every device to allows users interaction is smooth.

2.3.4 Maintainability

- Updating the app should be hard.
- The code base should be well structured and easily understandable
- Fixing bugs should be viable
- The code should also be well documented.

2.3.5 Scalability

- The app should be able to incorporate changes and should be able to handle new language courses added in the future

- Cultural scenarios can be updated and app should be handle them

2.3.6 Compatibility

- The app should be able to run on different platforms.
- The app must run on popular OS (Android).

2.3.7 User Feedback

- The user should be able to give feedback on how to improve the app.
- The user should be able to report any bugs they encounter.

2.3.8 Environmental Impact

- The app should be developed in a way that it consumes low battery power and is environment friendly.

2.4 Domain Model

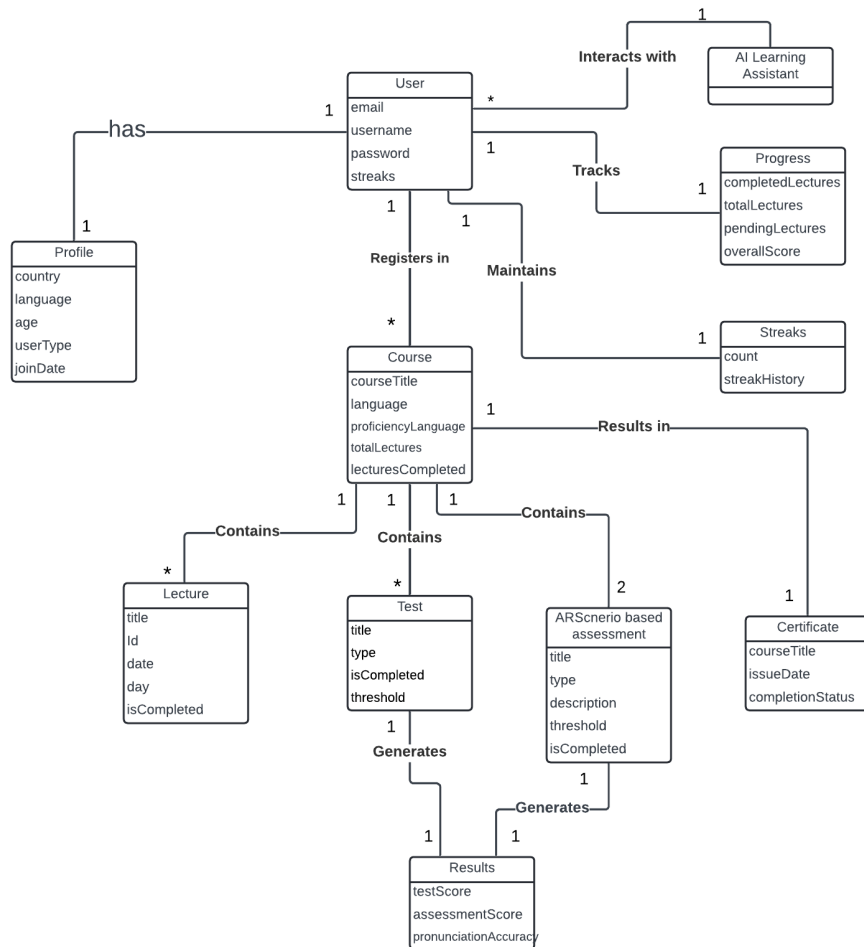


Figure 2.2: Domain Model

Chapter 3

System Overview

3.1 Architectural Design

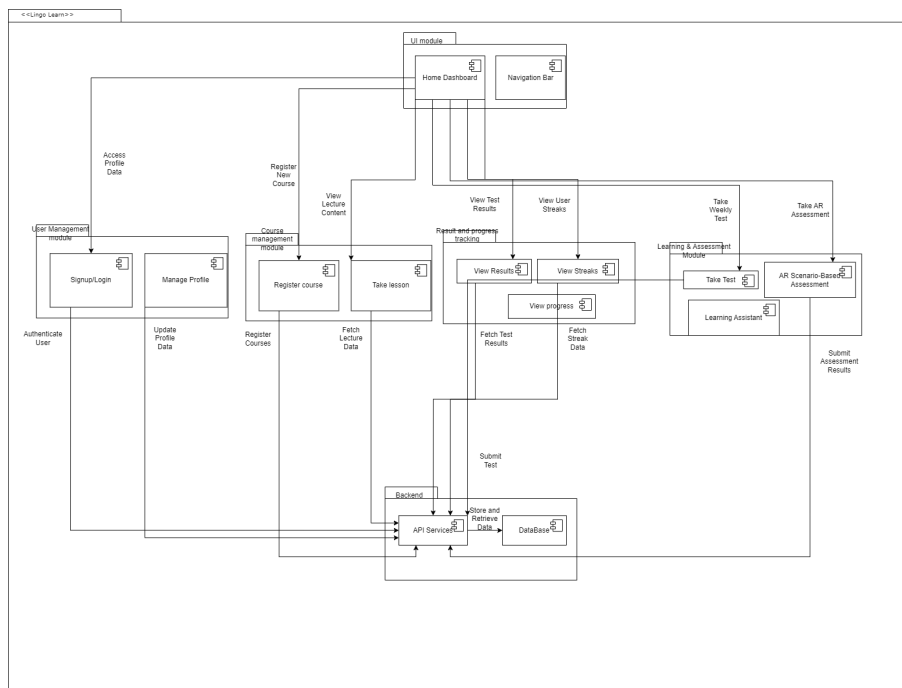


Figure 3.1: Architectural Design Diagram

3.2 Design Models

3.2.1 Data Flow Diagrams

Level 0:

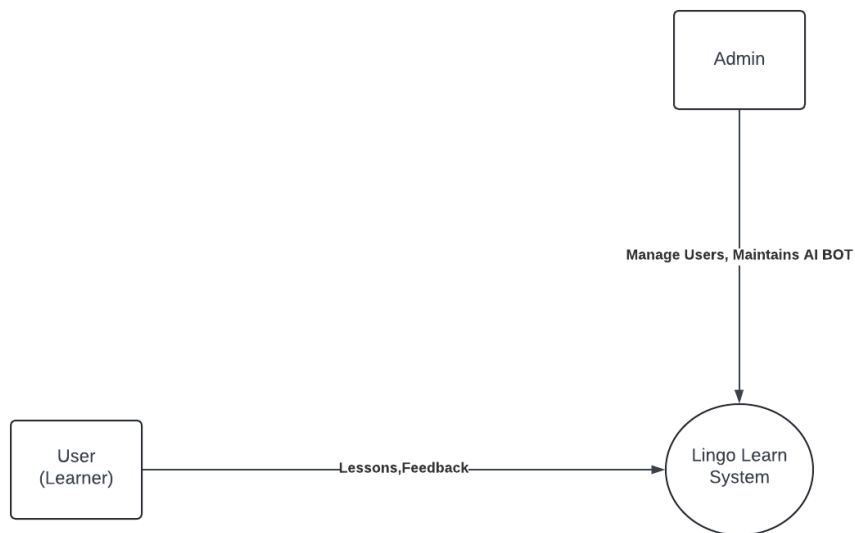


Figure 3.2: Level 0: Context Diagram

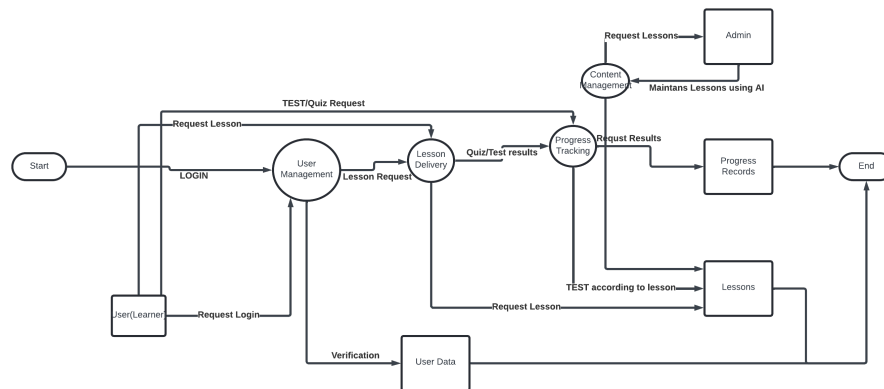
Level 1:

Figure 3.3: Level 1 Diagram

Level 2:

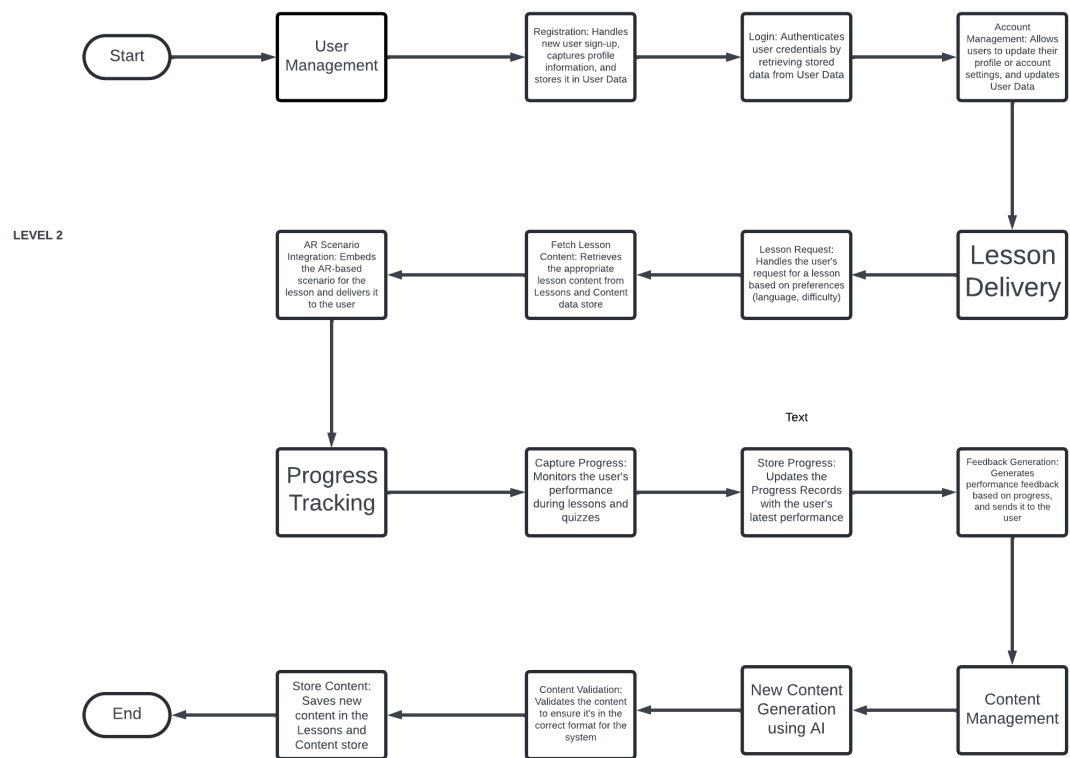


Figure 3.4: Level 2 Diagram

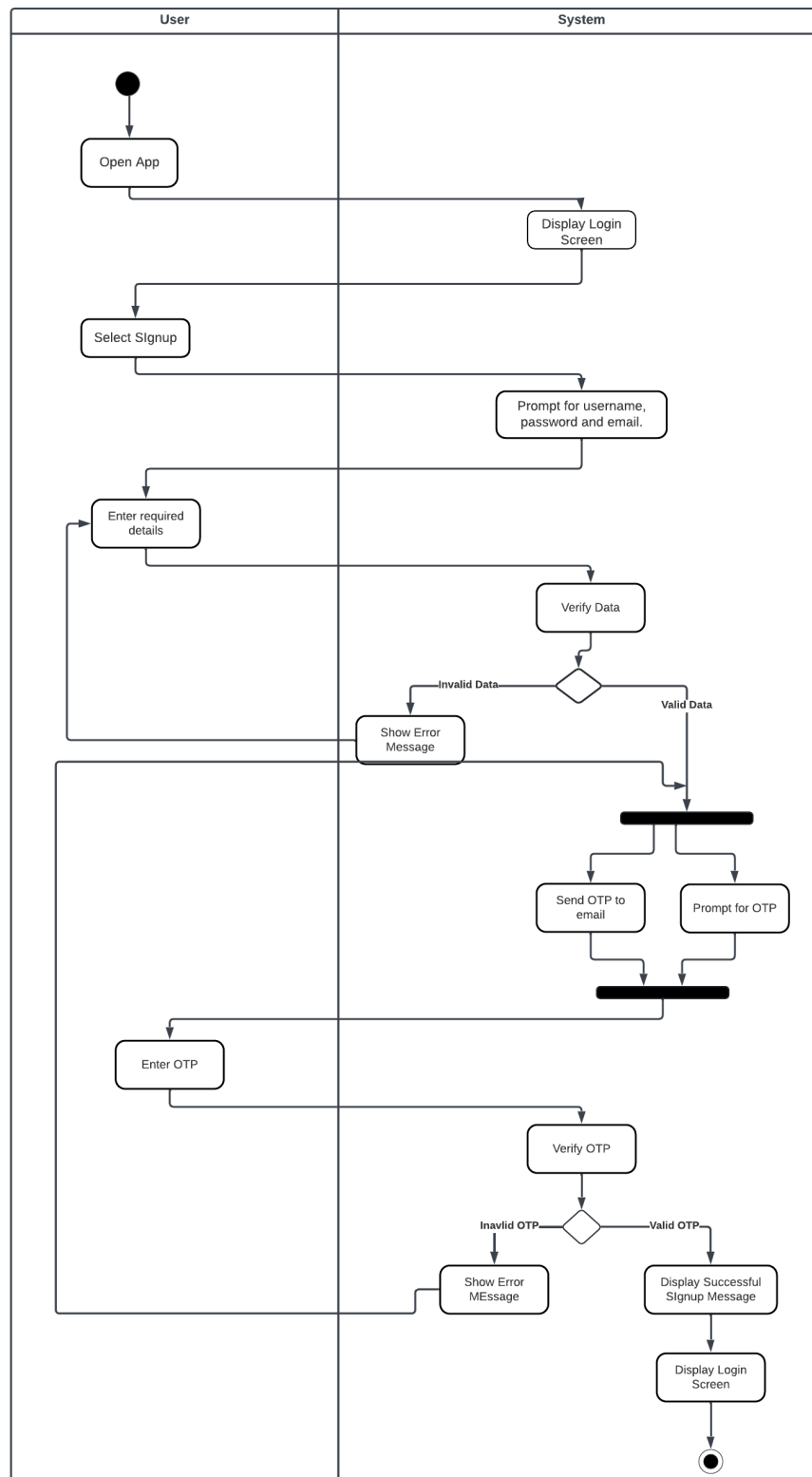
Activity Diagrams:

Figure 3.5: Activity Diagram 1:Sign Up

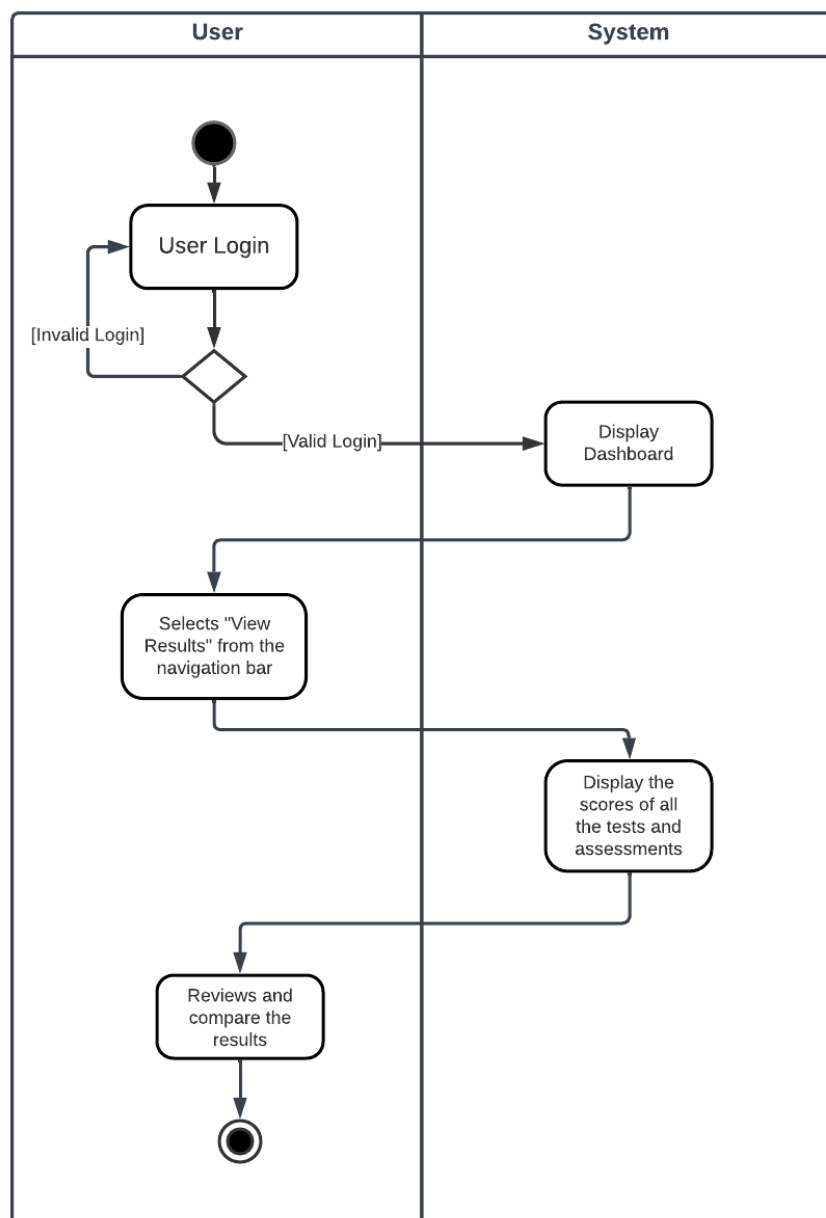


Figure 3.6: Activity Diagram 2:View Results

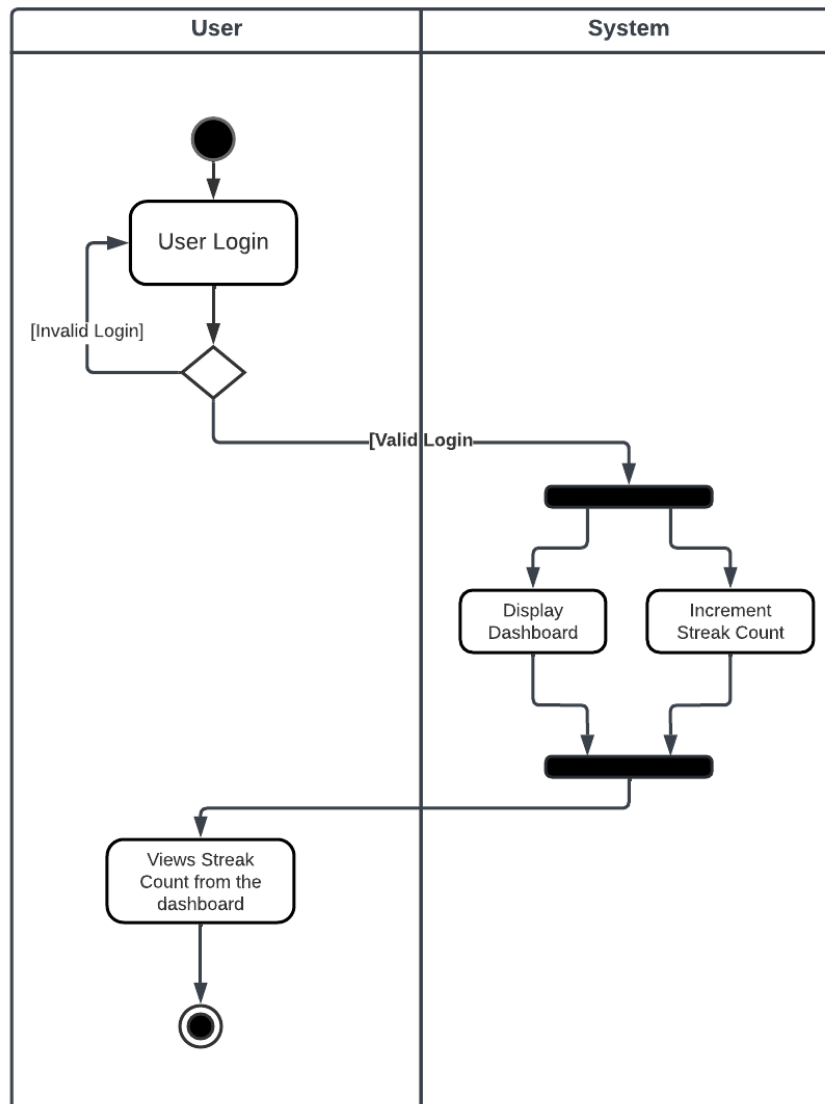


Figure 3.7: Activity Diagram 3:View Streaks

Take Lecture:

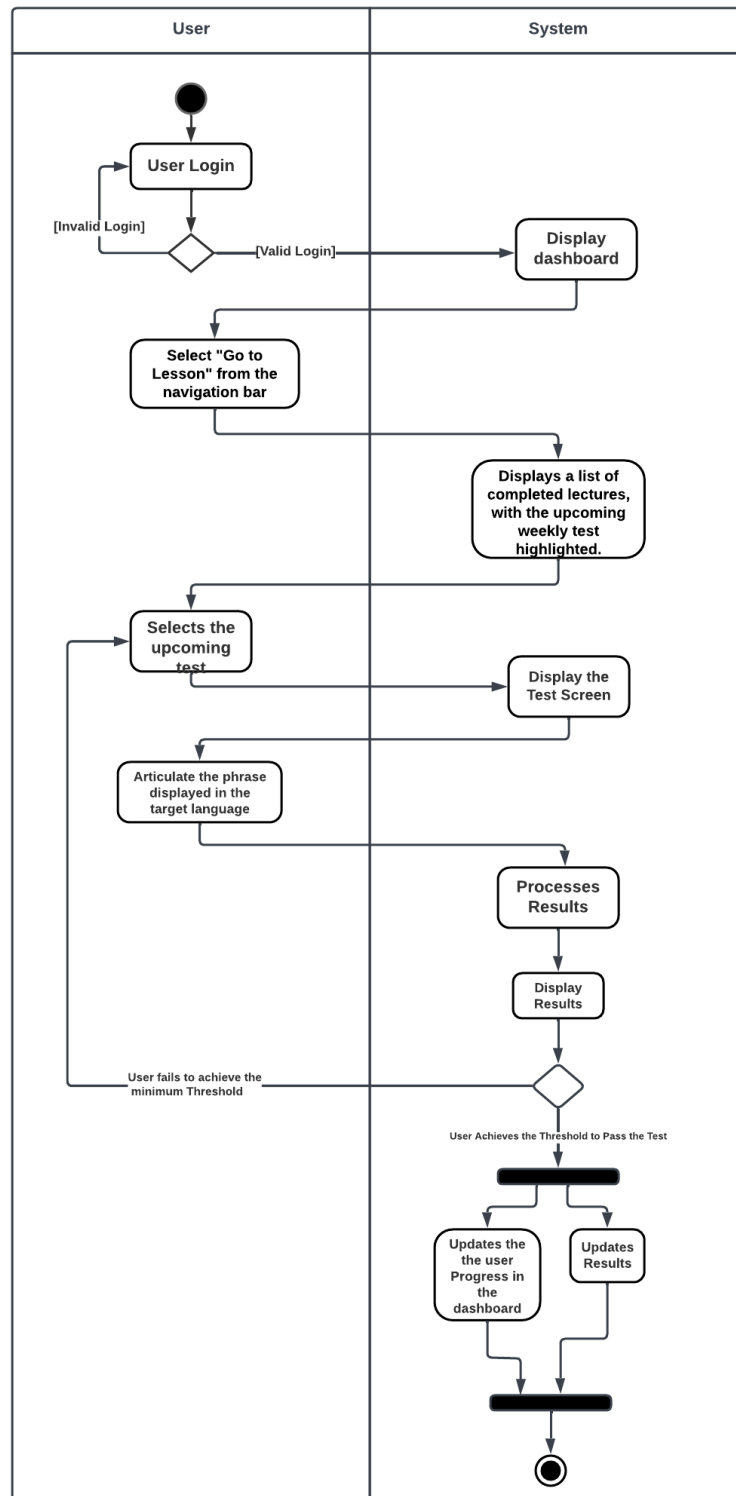


Figure 3.8: Activity Diagram 4:Take test

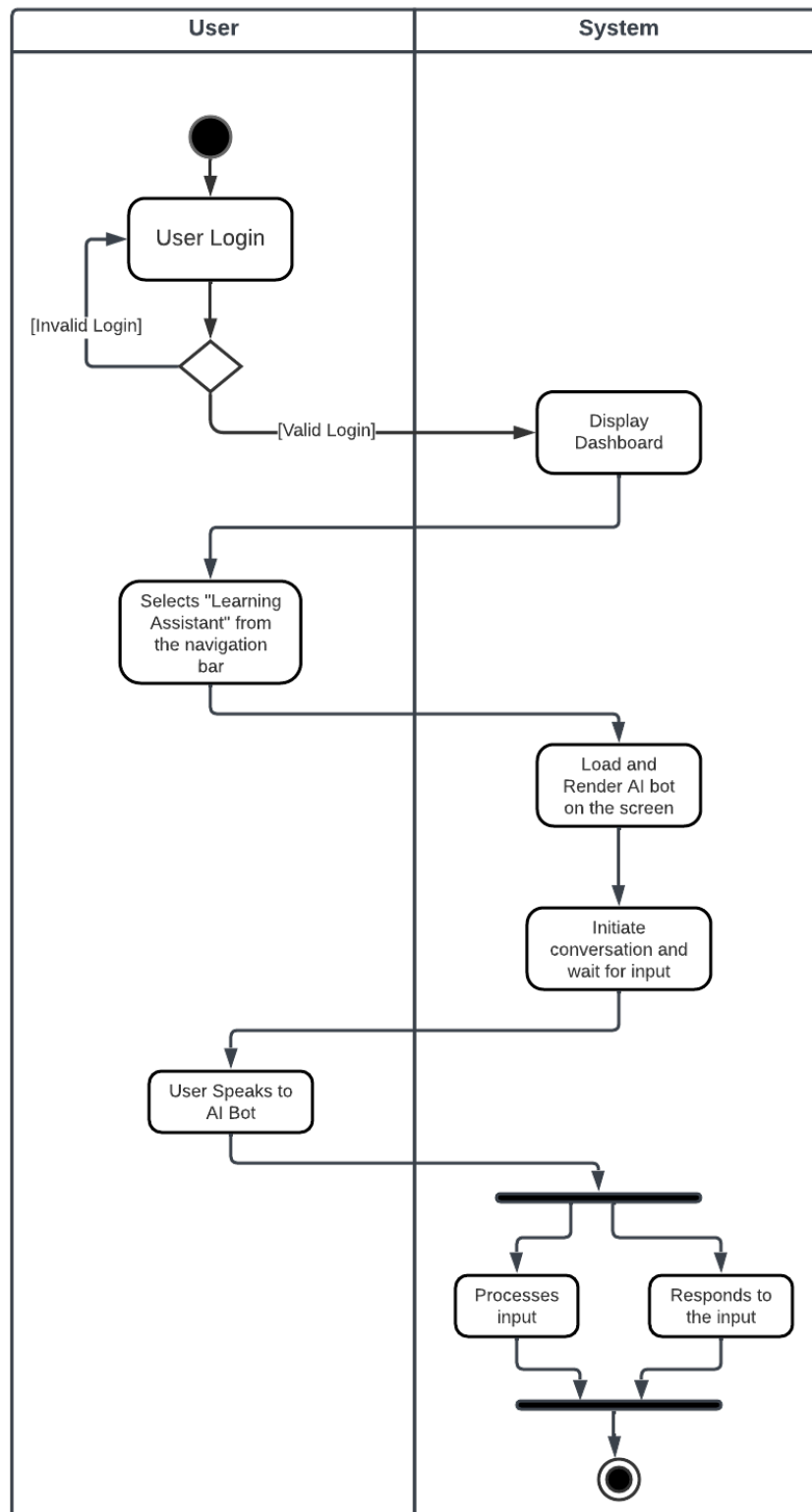
Interact with learning Assistant:

Figure 3.9: Activity Diagram 5:Interact with Learning Assistant

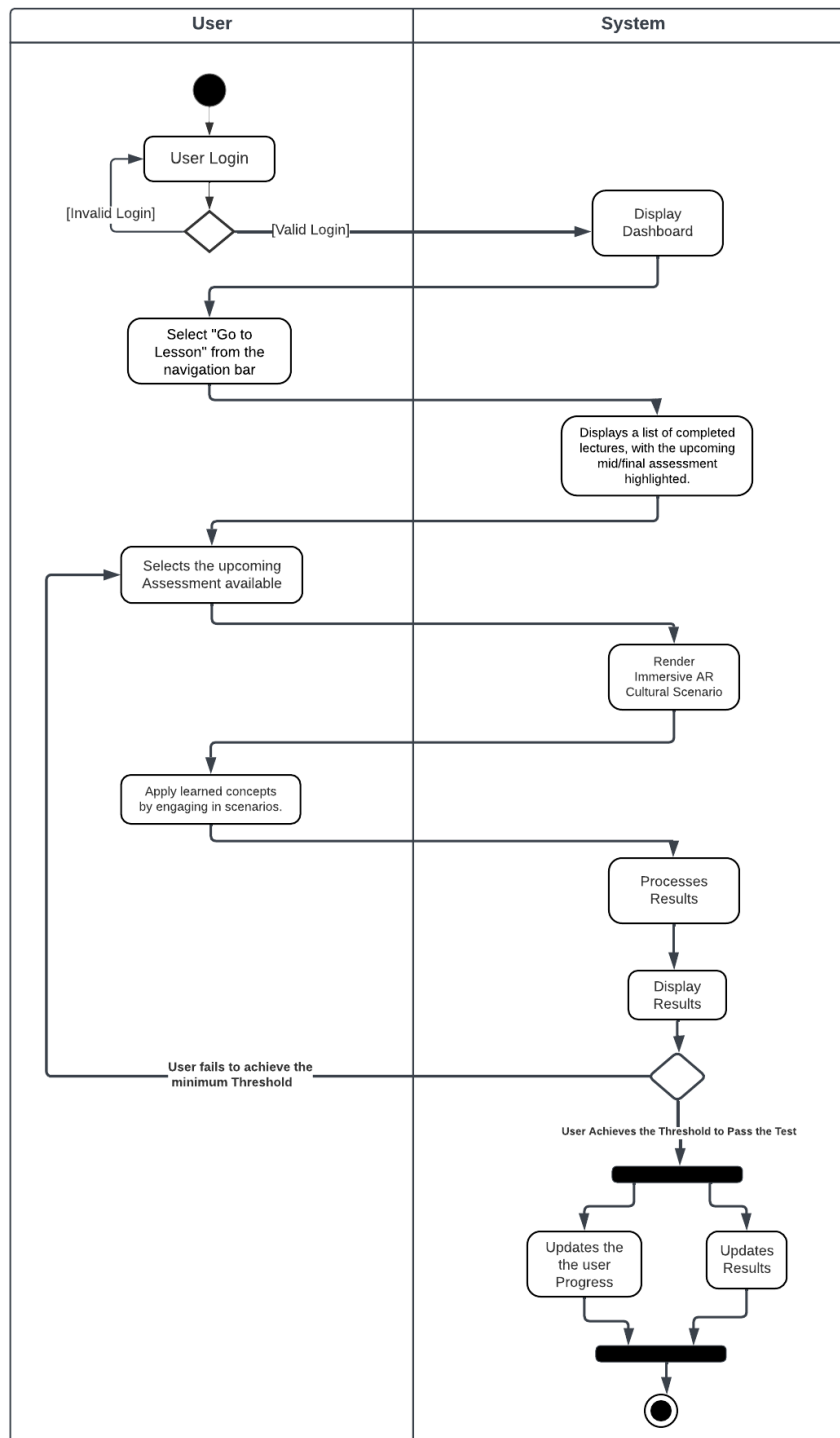
Take AR Scenerio-Based Assessment:

Figure 3.10: Activity Diagram 6:Take AR Scenario-Based Assessment

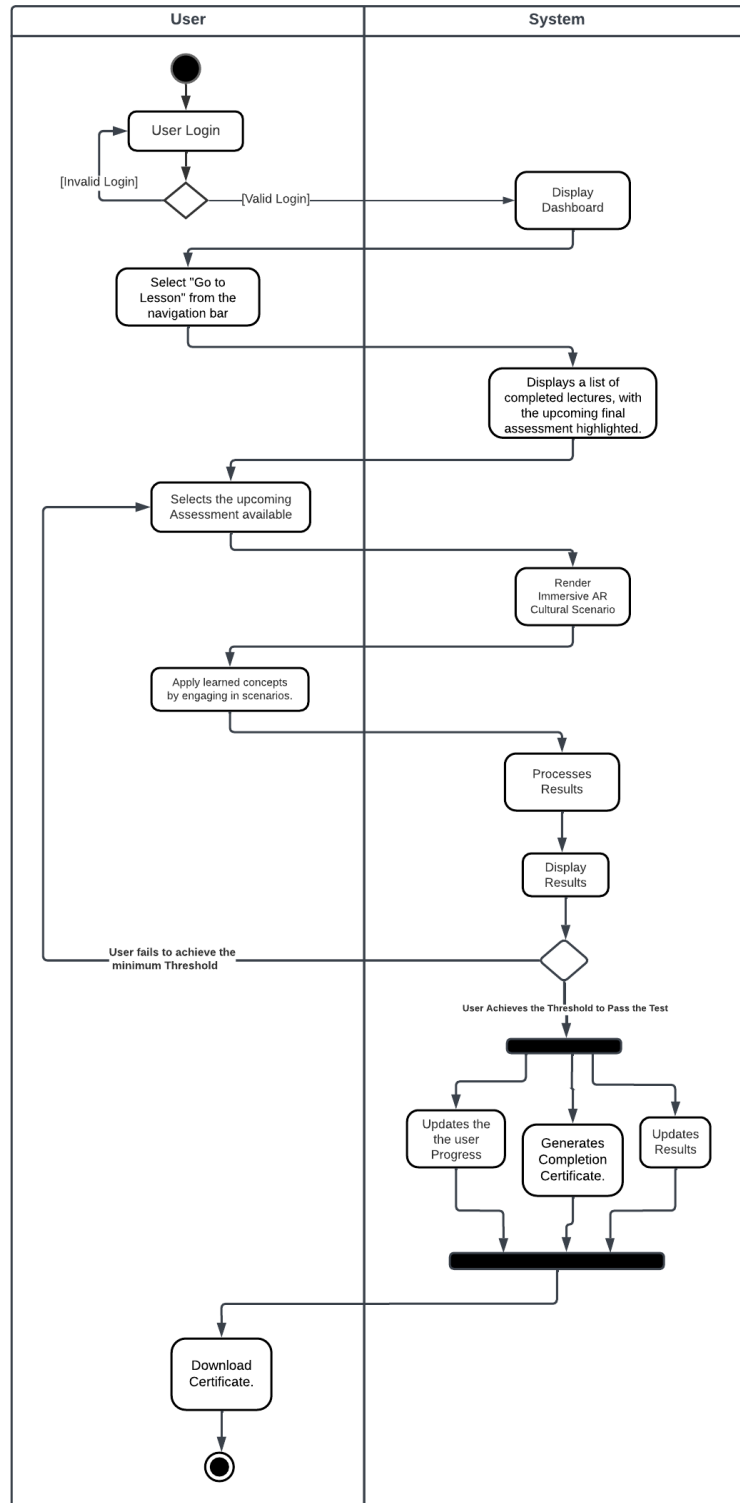
Generate Certificate:

Figure 3.11: Activity Diagram 7:Generate Certificate

Take Lecture:

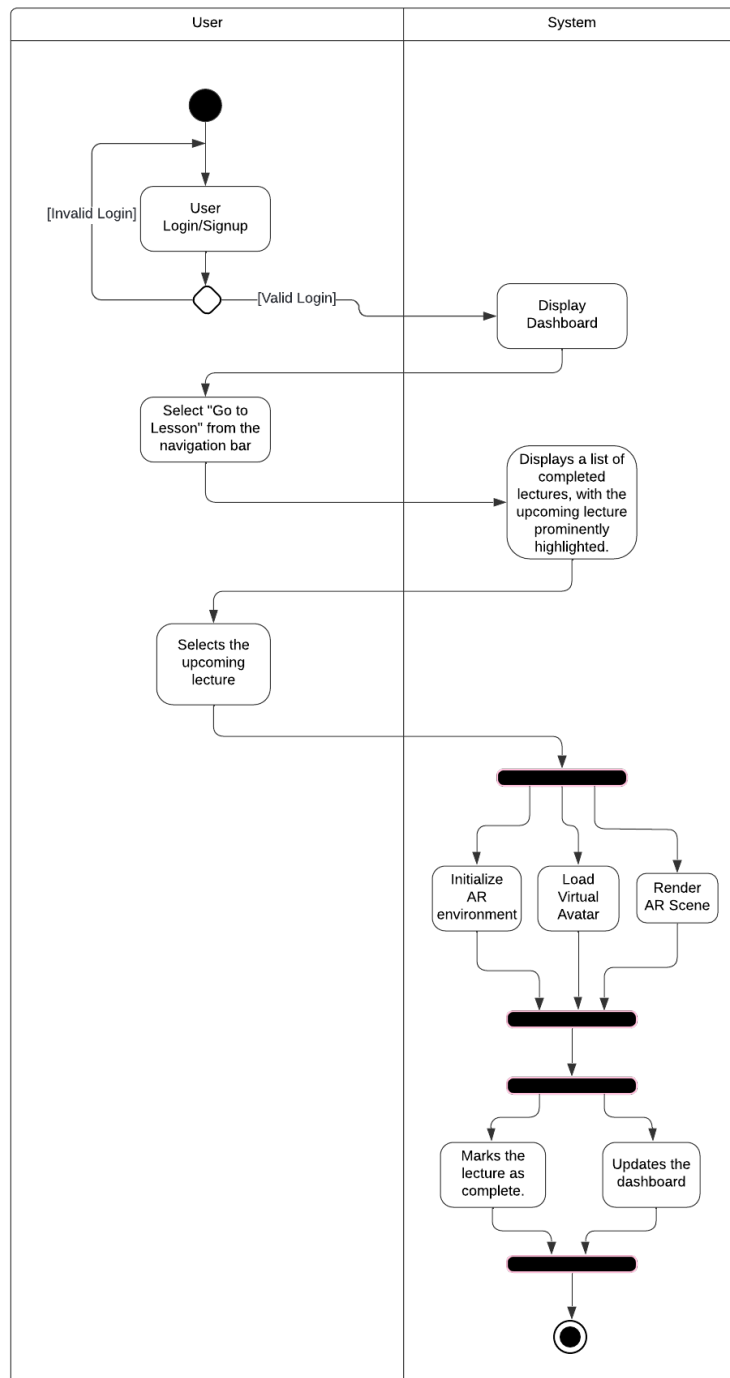


Figure 3.12: Activity Diagram 8:take lecture

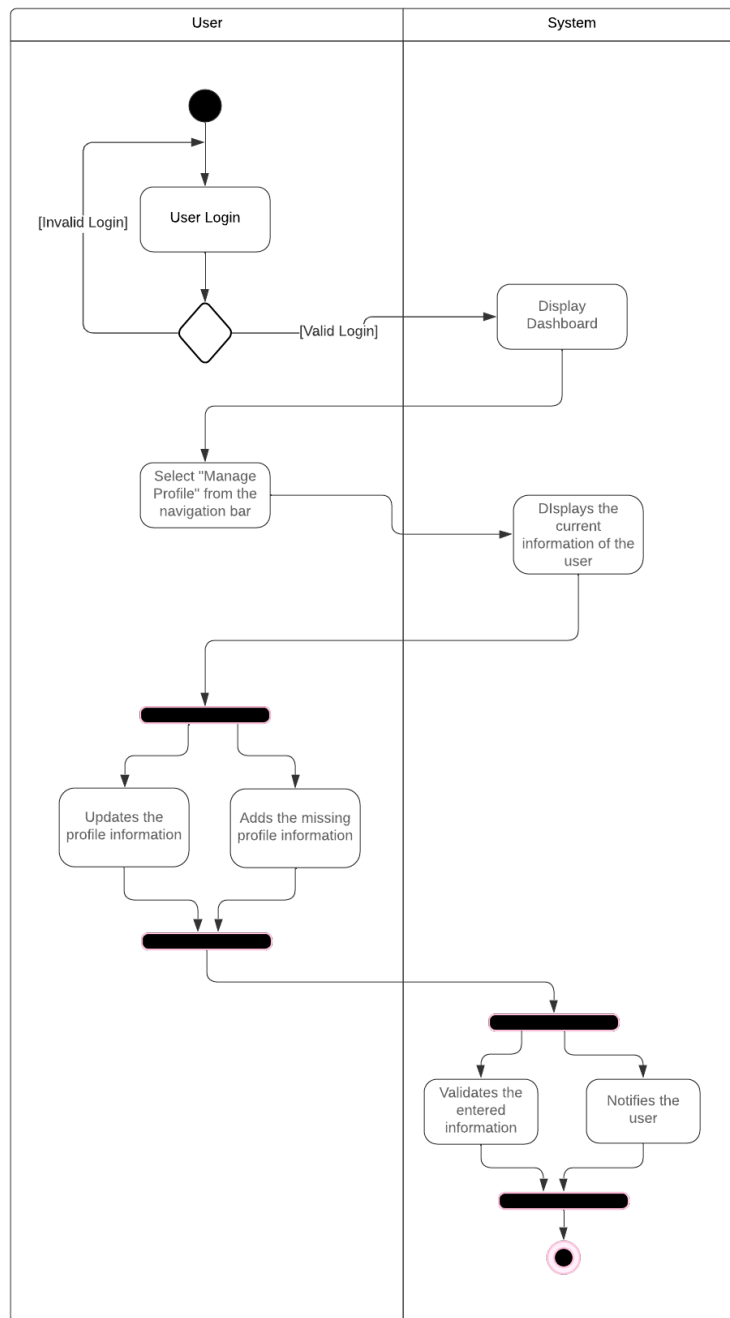
Manage Profile:

Figure 3.13: Activity Diagram 9:Manage Profile

Register Course:

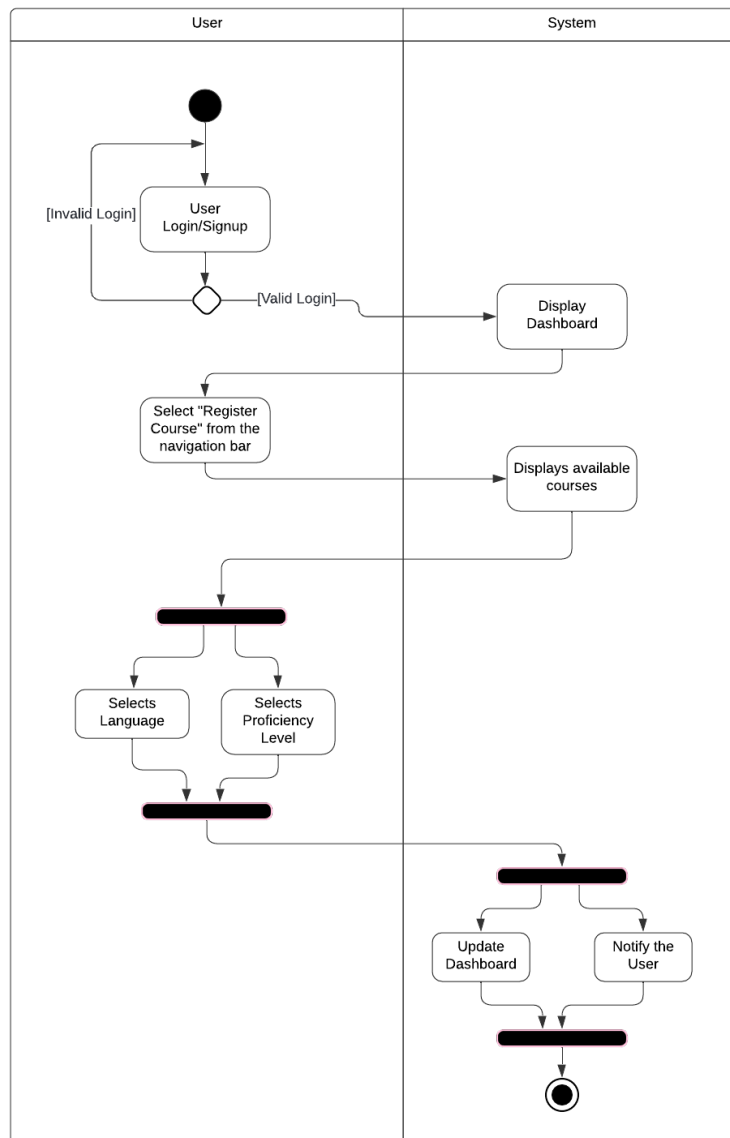


Figure 3.14: Activity Diagram 10:Register Course

3.2.2 System Level Sequence Diagrams:

Sign up:

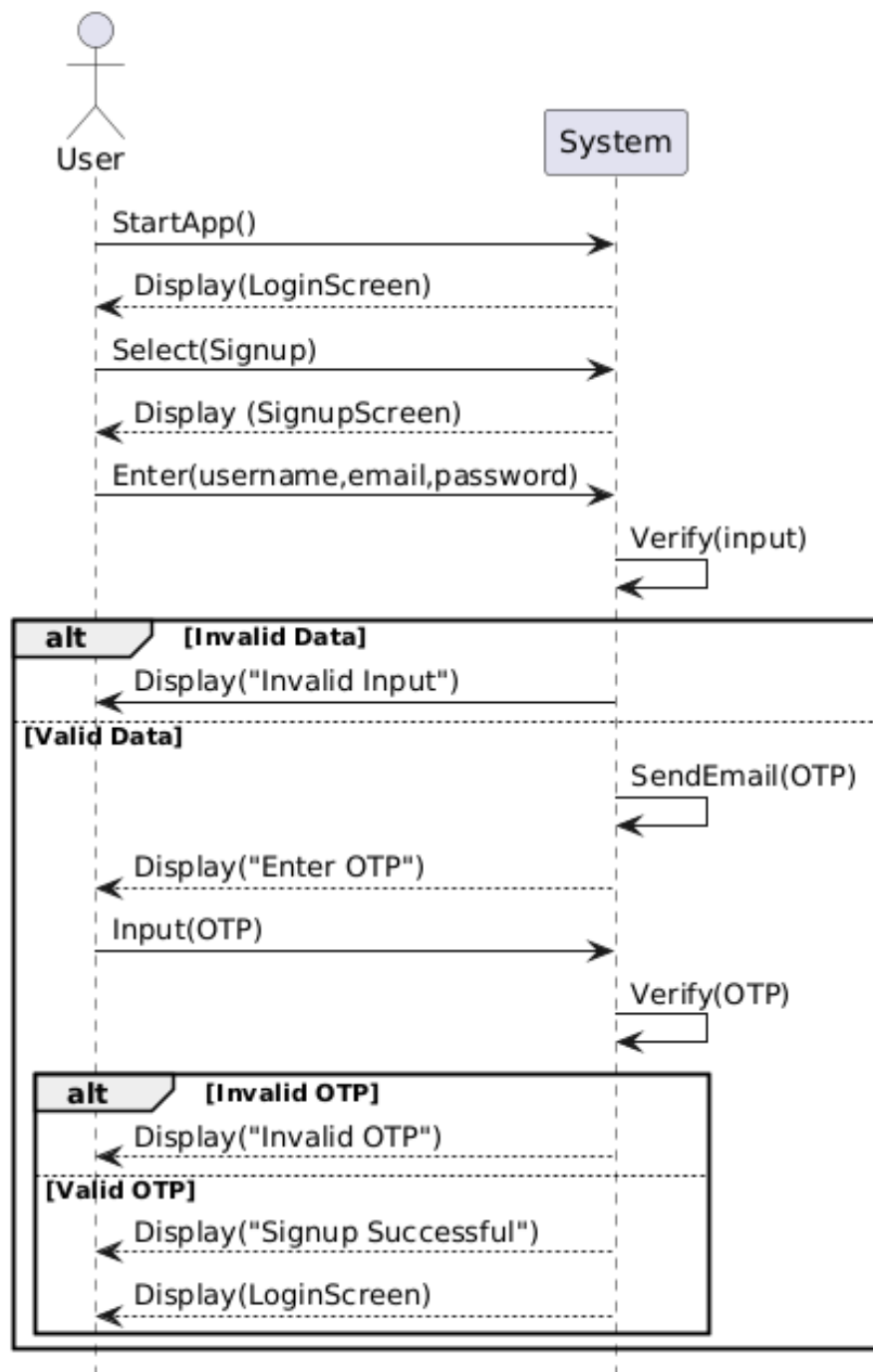


Figure 3.15: SSD 1:Sign Up

Manage Profile:

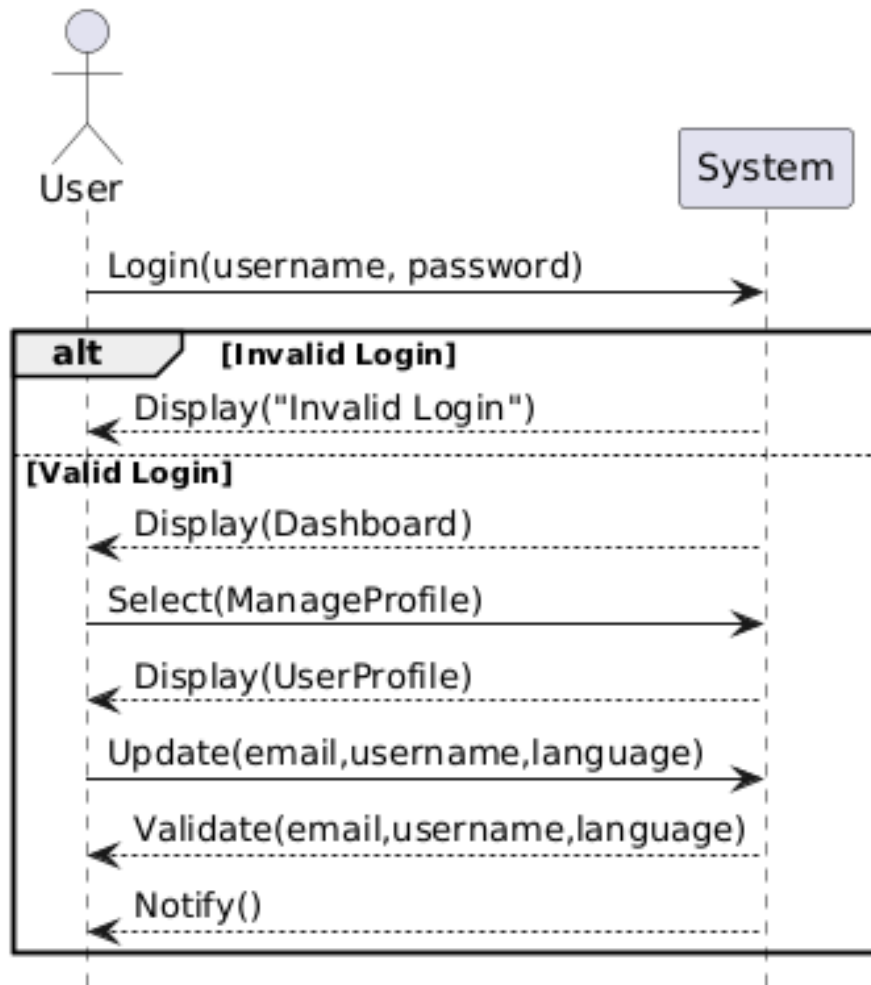


Figure 3.16: SSD 2:Manage Profile

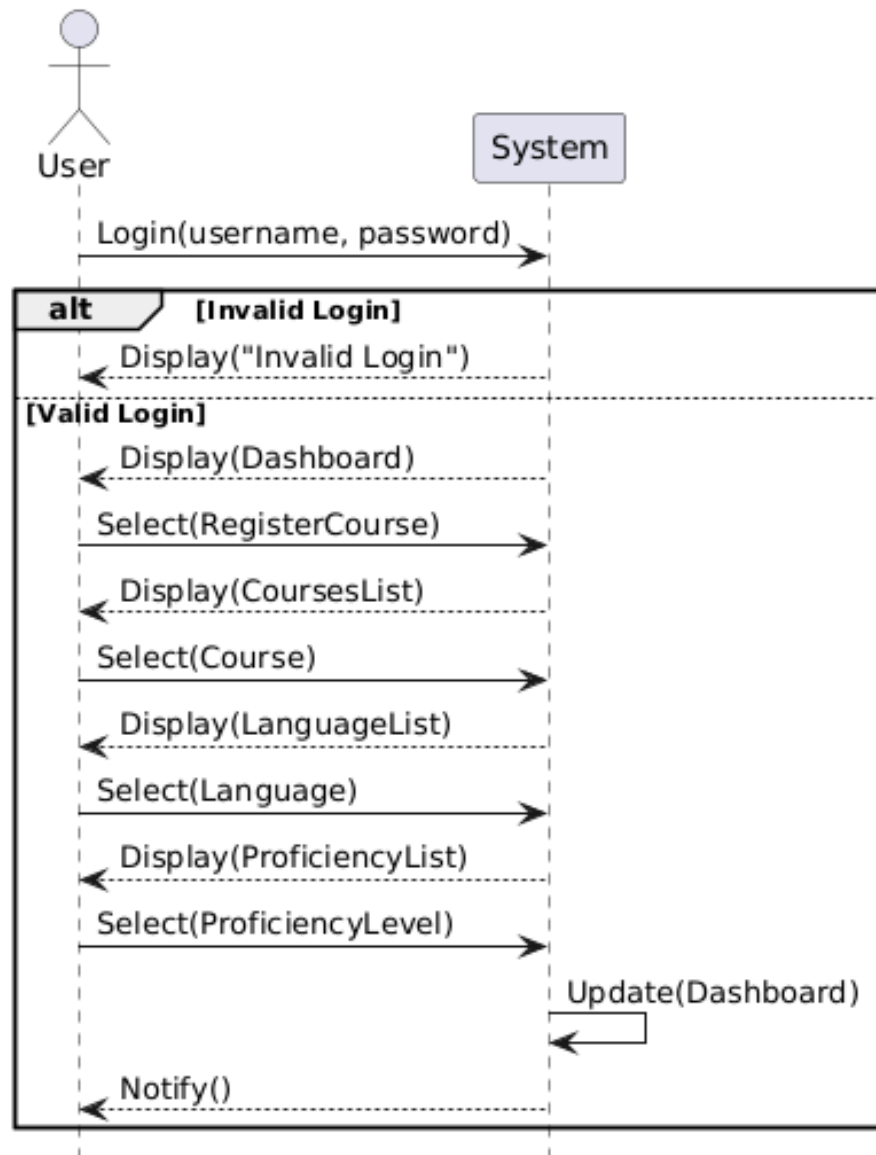
Register Course:

Figure 3.17: SSD 3:Register Course

Take Lecture:

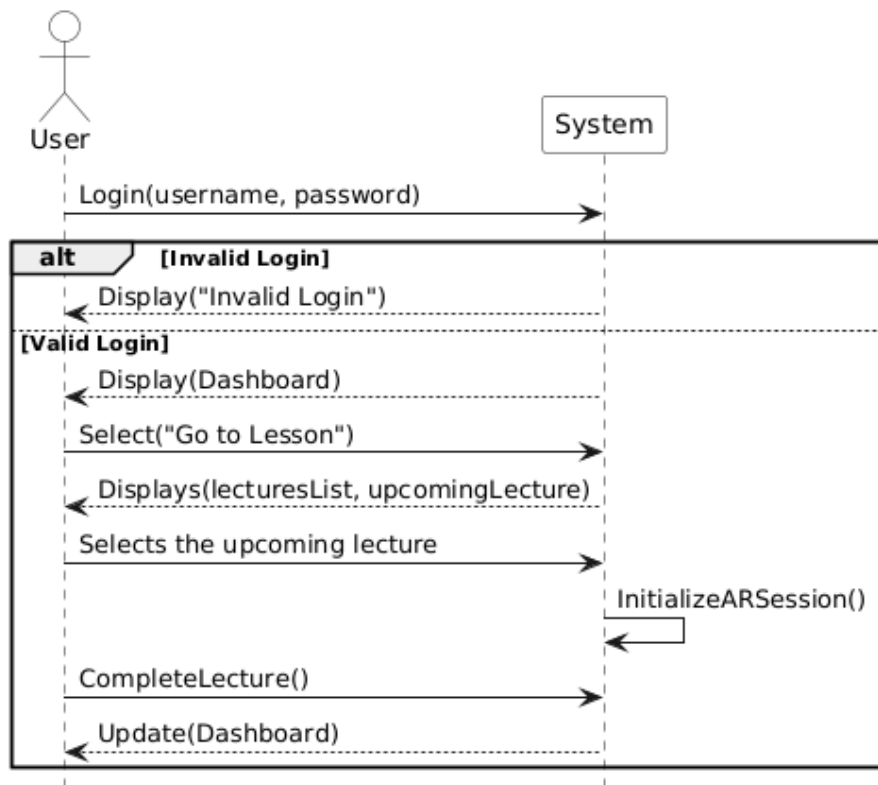


Figure 3.18: SSD 4:Take Lecture

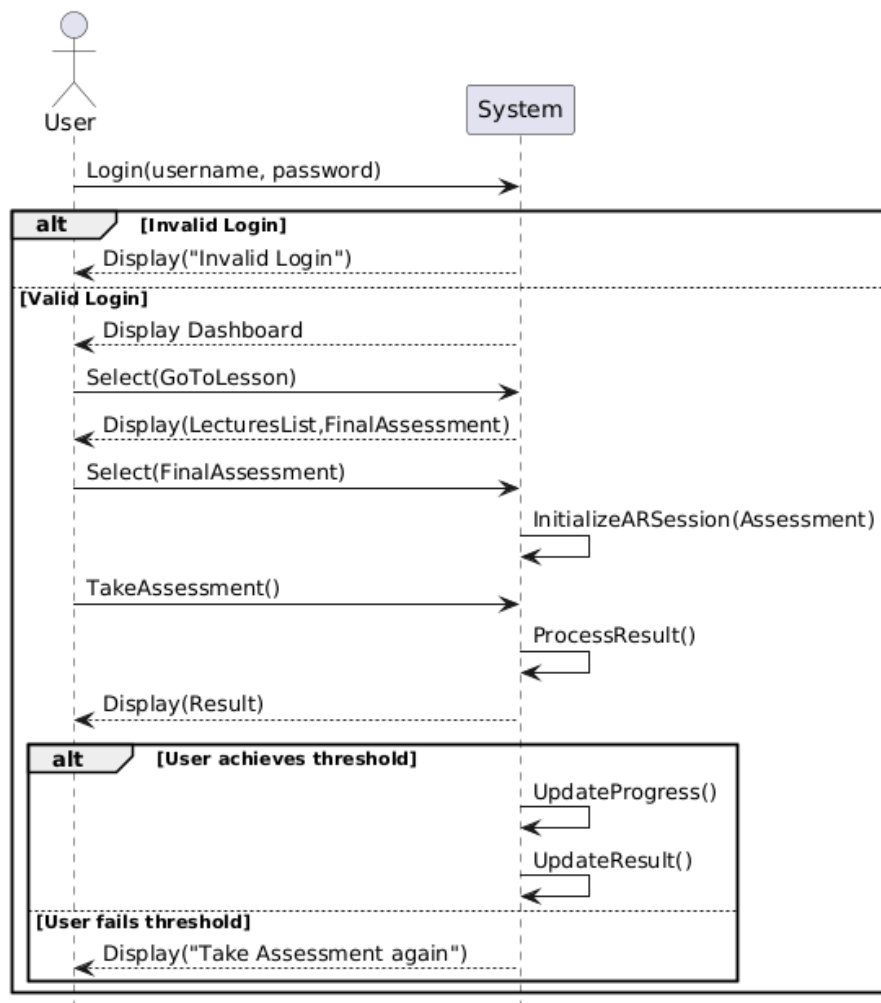
Take Test:

Figure 3.19: SSD 5:Take Test

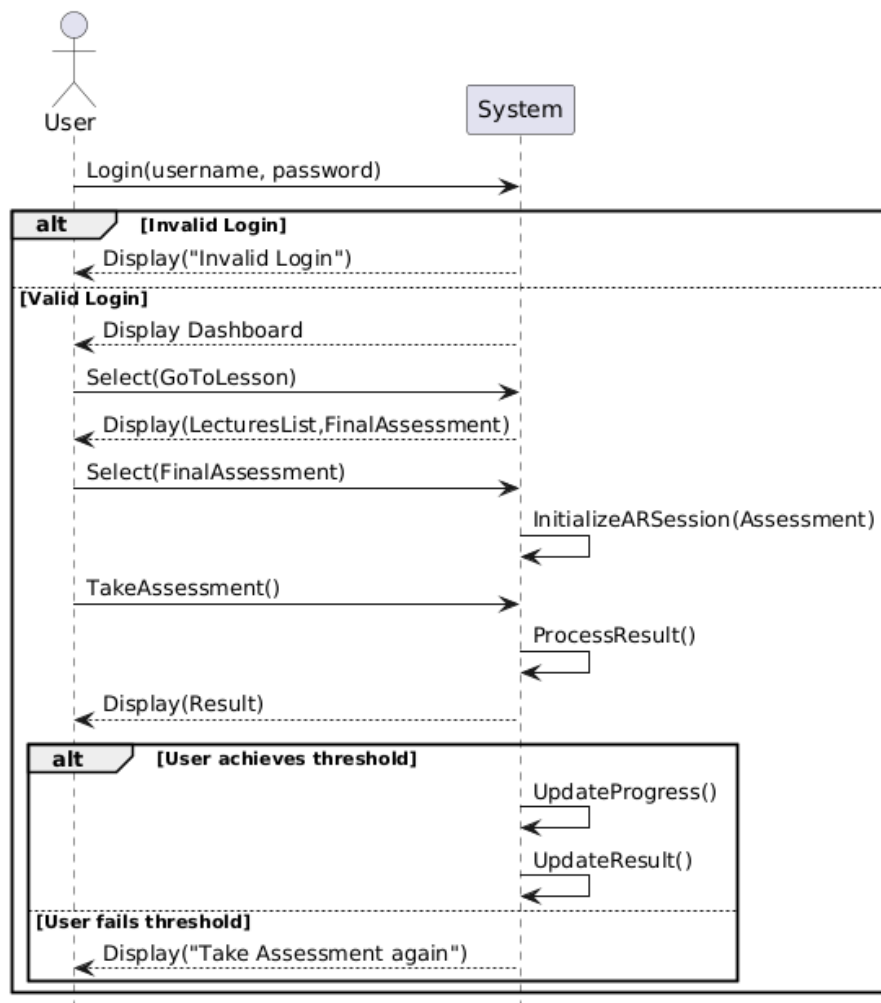
Take Assessment:

Figure 3.20: SSD 6:Take Assessment

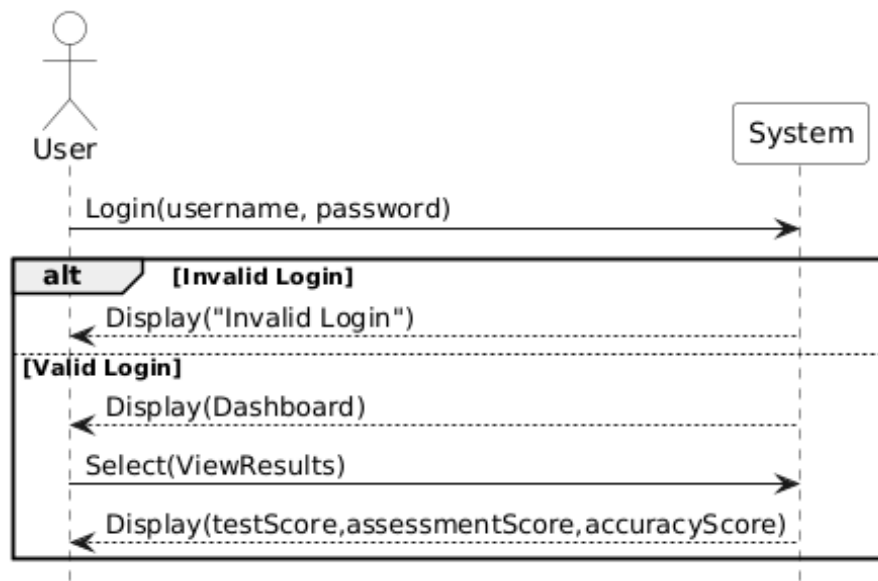
View Results:

Figure 3.21: SSD 7:View results

View Streaks:

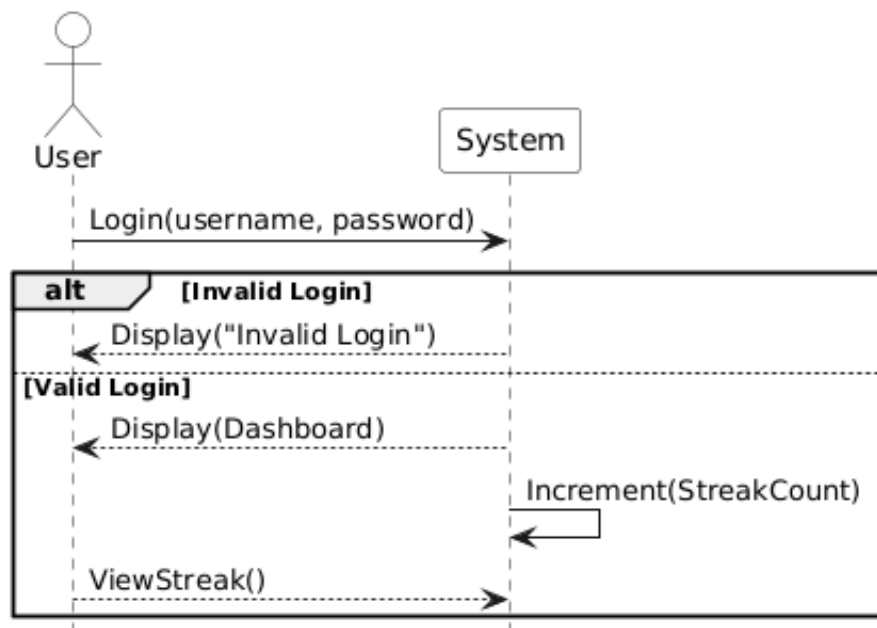


Figure 3.22: SSD 8:View Streaks

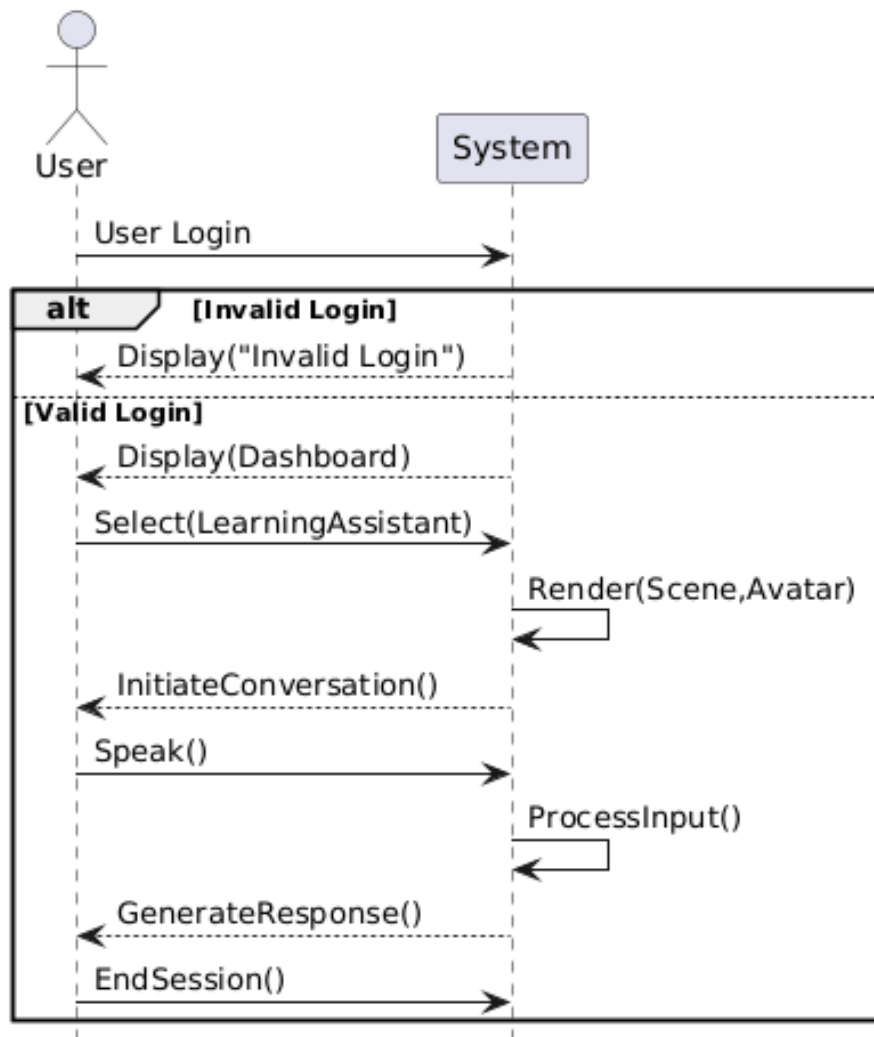
Interact with AI Assistant :

Figure 3.23: SSD 9:Interact with AI Assistant

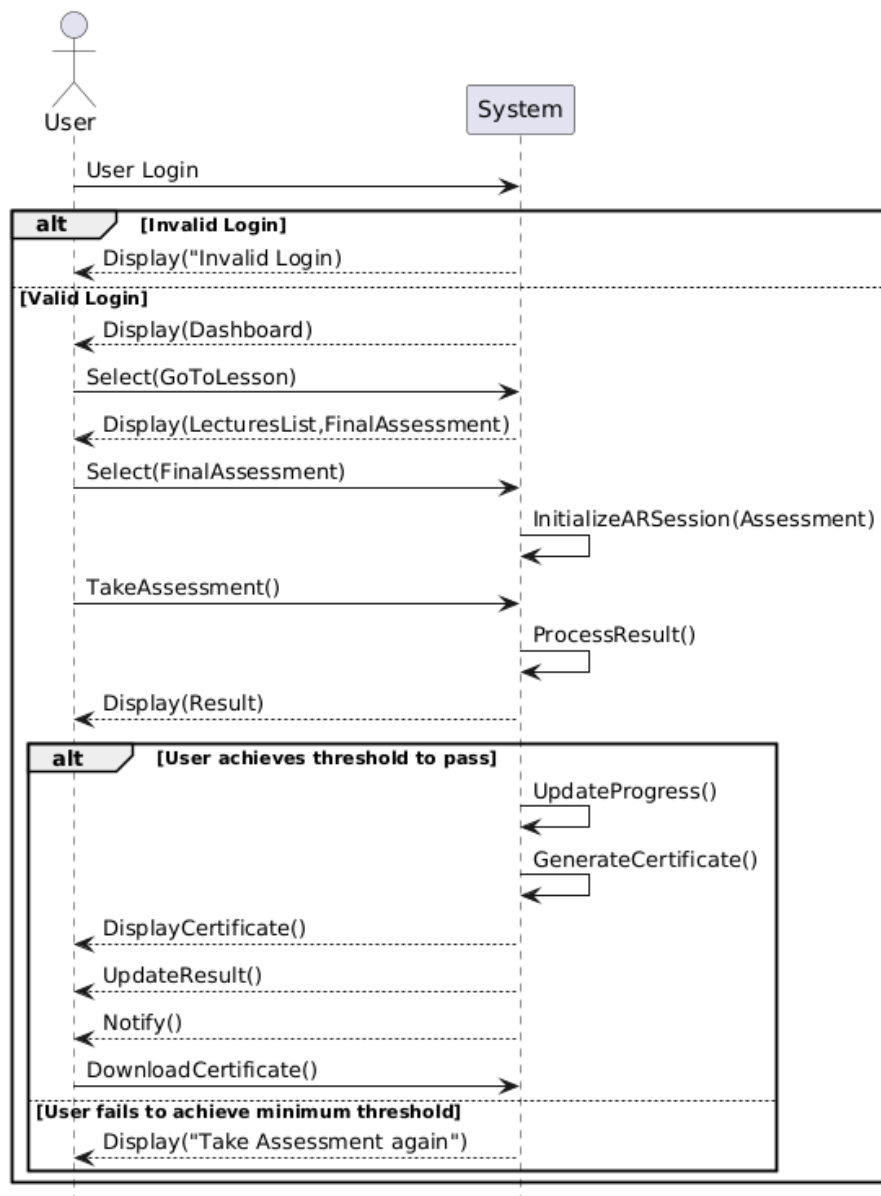
Generate Certificate:

Figure 3.24: SSD 10:Generate Certificate

3.2.3 State Transition Diagram

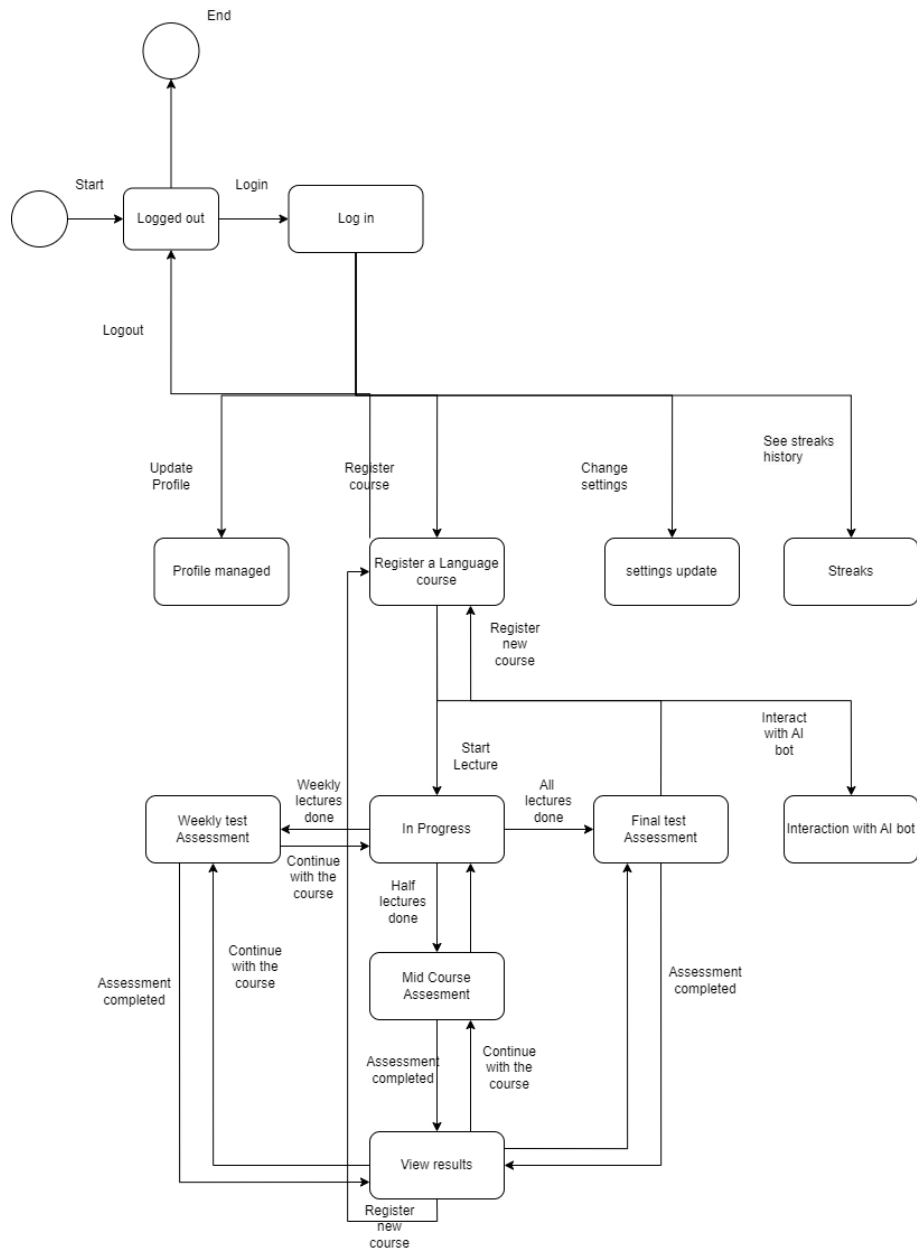


Figure 3.25: State Transition Diagram

3.3 Data Design

3.3.1 Entity Relationship Diagram:

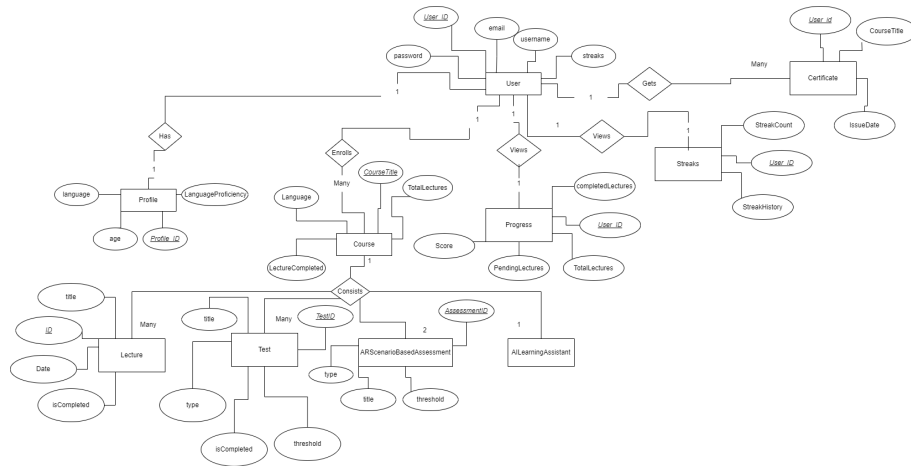


Figure 3.26: Entity Relationship Diagram

Bibliography

- [1] Anon. Nine-year old girl dies due to language barrier, interpreting absence at hospital—stratus video says lack of industry standardization creating healthcare hazard. *www.healthitoutcomes.com*, 2014.
- [2] Adam Hoffman. Millions of americans are getting lost in translation during hospital visits. *Smithsonian Magazine*, 2015.